

1.Experiment: FIND-S Algorithm

To implement and demonstrate the **FIND-S algorithm** for finding the **most specific hypothesis** using training data.

Python Program

```
import pandas as pd

from google.colab import files

uploaded = files.upload()

filename = list(uploaded.keys())[0]

df = pd.read_csv(filename)

print("Training Data:\n")

print(df)

hypothesis = None

for i in range(len(df)):

    if df.iloc[i, -1] == "Yes":

        if hypothesis is None:

            hypothesis = list(df.iloc[i, :-1])

        else:

            for j in range(len(hypothesis)):

                if hypothesis[j] != df.iloc[i, j]:

                    hypothesis[j] = "?"

print("\nFinal Most Specific Hypothesis:")
```

```
print(hypothesis)
```

Output

Final Most Specific Hypothesis:

```
['?', '?', '?', '?', '?', 'Same', '?', 'Outdoor', '?']
```